

Note for section 6

Ruihua Fan¹

¹*Department of Physics, Harvard University, Cambridge MA 02138, USA*

March 22, 2026

Abstract

In this note, we discuss Kitaev's honeycomb model. We will follow [1, 2].

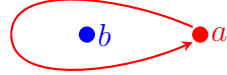
Contents

1	Introduction	2
2	The model	3
3	Perturbative analysis: Abelian anyons	5
3.1	Perturbation calculation, effective Hamiltonian	5
3.2	Abelian anyons	6
3.3	Weak symmetry breaking and gapless edge	7
4	Parton construction and the full phase diagram	10
4.1	Majorana fermion representation	10
4.2	The spectrum and the phase diagram	12
5	Magnetic field and non-Abelian anyons	15
5.1	Effective Hamiltonian and the spectral Chern number	16
5.2	Edge mode, thermal transport and unpaired Majorana fermion	16
5.3	Fusion and braiding	18
A	Perturbation in the Green's function formalism	18
B	Another fermion parton construction	20
B.1	Slave fermion: generality	20
B.2	Examples	20
C	Review of $p + ip$ superconductor	20

1 Introduction

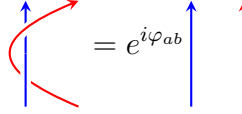
Anyons are particles with unusual statistics. In three-dimensional gapped systems, boson and fermion are the only options.¹ In two-dimensional gapped systems, there can be alternatives, which are called anyons. There are two types of them, abelian and non-abelian anyons:

- For abelian anyons, the wavefunction $|\Psi\rangle$ will acquire a phase φ_{ab} if a particle a moving around another particle b counterclockwisely. The phase only depend on the type of two particles not on the path that a chooses, i.e.



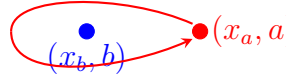
$$|\Psi\rangle \mapsto e^{i\varphi_{ab}} |\Psi\rangle \quad (1)$$

It is more convenient to represent the above equation using the following spacetime diagram, where the time goes upward



$$= e^{i\varphi_{ab}} \quad (2)$$

- For non-abelian anyons, even if we just put several of them on the plane, the state will have degeneracy. Let $\mathcal{L}(x_a, a; x_b, b)$ denote the degenerate subspace, which depends on the anyon types and also their positions. The state will be transformed by a unitary matrix when one particle encircles the other, i.e.



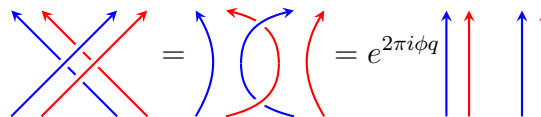
$$|\Psi\rangle \mapsto U |\Psi\rangle, |\Psi\rangle \in \mathcal{L}(x_a, a; x_b, b) \quad (3)$$

One simple realization of anyon is the 2+1D $U(1)$ gauge theory. It contains magnetic vortex $\phi \in \mathbb{R}/\mathbb{Z}$ and electric charge $q \in \mathbb{Z}$. When an electric charge moves around a magnetic vortex, the system acquires an Aharonov–Bohm phase



$$|\Psi\rangle \mapsto e^{2\pi i q \phi} |\Psi\rangle \quad (4)$$

We can also consider a dyon (q, ϕ) , which is the composite particle q of an electric charge and a magnetic vortex ϕ . It is interesting to ask what the self-statistic of a dyon looks like, which amount to calculating the phase generated by exchanging two dyons. By using the spacetime diagram, we have



$$= e^{2\pi i \phi q} \quad (5)$$

¹For gapless systems, the phase or generally the transfer matrix may depend on the path and it is tricky to talk about statistics in the way that we will use below.

In the second step, we use the property that charge and vortex are boson among themselves to remove some crossing. In the last step, we use the mutual statistics between charge and vortex.

The goal is to find anyons in real materials. Although the model that we write down may still look artificial, it is much more realistic than having gauge fields to start with. Let us list the rules of game:

- Spins on a lattice. The Hilbert space has a simple tensor product structure $\mathcal{H} = \otimes_x \mathcal{H}_x$, where $\mathcal{H}_x$ denotes the Hilbert space of each site.
- Local Hamiltonian. Note that we impose locality on the Hamiltonian not on the Lagrangian. This is because sometimes local Lagrangian can translate to a non-local Hamiltonian (due to anomaly). And it is more natural to put constraints on the Hamiltonian for our purpose.
- Interested mostly in gapped phases. Gapless systems may be more interesting but also more complicated, the particle statistics may not be well-defined either.

Let us provide a clearer explanation of this. In a gapped system, spin correlation functions decay exponentially with distance. Accordingly, two quasi-particles cannot interact directly because any local action on one particle does not influence the other. As a result, they can only interact topologically if they move around each other. This is exactly the defining features of anyons, the *braiding rules*. [in gauge theory, do we need a gap for anyon to be well-defined? it seems that it is always sensible to talk about the AB phase]

- No exact symmetries are assumed. Because we want the properties that we are interested in to be robust against a generic local perturbation. Note that emergent symmetries in the IR are certainly allowed. We are also allowed to introduce gauge symmetries explicitly in the UV by adding fictitious degrees of freedom into the system

The toric code model is one example. But it involves four-spin interaction. Models with less-spin interaction may be more realistic and easier to realize in the real world. In this note, we will study a model proposed by Kitaev, which consists of only two-spin interaction, and see how it can realize both abelian and non-abelian anyons in different parameter regime. A salient feature of this model is that it can be solved exactly.

2 The model

We study a spin-1/2 system on a honeycomb lattice, where the spins are attached to the vertices, see Fig. 1. Each unit cell consists of two sublattices, denoted by the red and blue dots. Links are divided into three types, depending on their directions; we call them x -links, y -links and z -links. The Hamiltonian is

$$H = -J_x \sum_{x\text{-link}} \sigma_j^x \sigma_k^x - J_y \sum_{y\text{-link}} \sigma_j^y \sigma_k^y - J_z \sum_{z\text{-link}} \sigma_j^z \sigma_k^z \quad (6)$$

where J_x, J_y, J_z are model parameters that are generally different.

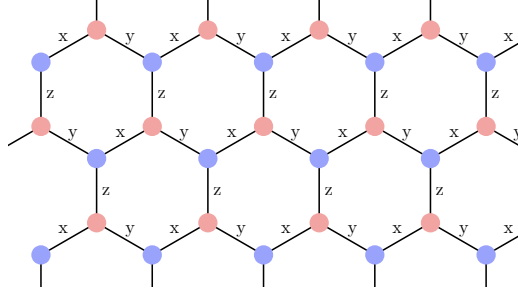


Figure 1: The honeycomb lattice for the Kitaev model. There are three types of links, each type is assigned to a different coupling.

We want to identify the symmetries in the model. There are two obvious symmetries, the translational invariance and time-reversal invariance $T\sigma^\alpha T^{-1} = -\sigma^\alpha$. Remarkably, there is also an extensive number of *local* symmetries. Let us introduce the following mutually commuting plaquette operators

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z$$

$$[W_p, W_{p'}] = 0, \quad [W_p, H] = 0, \quad W_p^2 = 1$$

(7)

They are Hermitian and actually commute with individual term in the Hamiltonian. If we denote the two-spin interaction in the Hamiltonian by K_{ij} , then the plaquette operator is exactly the product of K_{ij} around the plaquette, i.e.,²

$$W_p = \prod_{(ij) \in \text{bdry}(p)} K_{ij} \quad (8)$$

The set of “integrals of motion” W_p greatly simplifies the problem. We can divide the full Hilbert space into sectors, each of which is labeled by the eigenvalues of W_p

$$\mathcal{H} = \bigoplus_{\{w\}} \mathcal{H}_{w_1, w_2, \dots, w_m} \quad (9)$$

where $w_j = \pm 1$ denotes the eigenvalue and m is the number of plaquettes. So a state $|\psi\rangle$ is in the sector $\mathcal{H}_{w_1, w_2, \dots, w_m}$ if and only if $B_p |\psi\rangle = w_p |\psi\rangle$. We would like to interpret w_p as the intensity of a $\mathbb{Z}_2$ magnetic vortex. Then we can represent the field strength in terms of $\mathbb{Z}_2$ vector potential $u_{jk} = \pm 1$ on the links, namely

$$w_p = \prod_{(jk) \in \text{bdry}(p)} u_{jk} \quad (10)$$

Determining u_{jk} for a given w_p is not unique and this creates a $\mathbb{Z}_2$ gauge symmetry. At this stage, it seems completely redundant to introduce such u_{jk} , however, we will see that they naturally appears later when we solve the model in the fermion representation.

²If the system is defined on a torus, the product of all plaquette operators is the identity. In addition, the product of K_{ij} along the non-contractible loops are also integrals of motion.

Now, let us understand what the integrals of motion buy us by a counting of degrees of freedom. Let n denote the number of vertices, then we roughly have $m \approx n/2$ for a large lattice. Since each plaquette operator gives a $\mathbb{Z}_2$ constraint, the number states within each sector $\mathcal{H}_{w_1, w_2, \dots, w_m}$ is $2^{n-m} \approx 2^{n/2}$. This means that the problem is not solved at this stage yet. More importantly, it provides a hint that the underlying degrees of freedom within each sector might be *Majorana fermions*, where the fermions are associated to vertices.

3 Perturbative analysis: Abelian anyons

In this section, we want to find parameter regimes that allow us to do perturbative analysis. Within the perturbation regime, we will show that the system can support abelian anyons and the anyon content can be identified with the toric code. More interestingly, it shows an example of weak symmetry breaking.

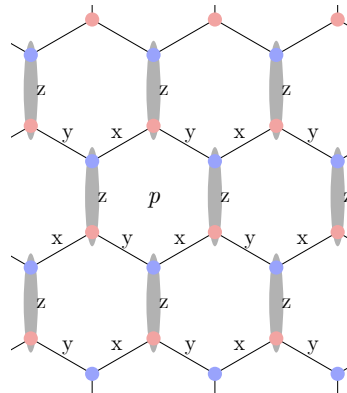
3.1 Perturbation calculation, effective Hamiltonian

Let us consider the limit $J_z \gg J_x, J_y$ and separate the Hamiltonian as follow

$$H_0 = -J_z \sum_{z\text{-link}} \sigma_j^z \sigma_k^z, \quad V = -J_x \sum_{x\text{-link}} \sigma_j^x \sigma_k^x - J_y \sum_{y\text{-link}} \sigma_j^y \sigma_k^y$$

Without loss of generality, we assume that $J_z > 0$ (the opposite case can be studied similarly).

We first solve the spectrum of H_0 . The ground state is highly degenerate: every two spins connected by a z -link are aligned in the z -direction. The excited states contain pairs of anti-aligned spins and each of them has an energy cost $2J_z$. We mainly focus on the ground state manifold and thus can effectively regard every such two spins as a big spin τ , shown below. The τ spins can be regarded as living on vertices of a square lattice. Each integral of motion W_p acts within the ground state manifold and can be effectively represented as a four- τ -spin term



$$W_p \rightarrow \tau_{\text{left}(p)}^y \tau_{\text{right}(p)}^y \tau_{\text{up}(p)}^z \tau_{\text{down}(p)}^z \quad (11)$$

Note that the number degenerate ground states is exactly the same as the number of integrals of motion, which implies that they can be further organized by the eigenvalues of W_p . Requiring the unperturbed states to be eigenstates of all W_p 's introduces non-local entanglement among the τ -spins. We will see that this requirement naturally appear in the perturbation calculation as an energetic condition.

Now, we include V perturbatively. We follow Appendix. A and define an effective Hamiltonian via the self-energy operator $\Sigma_-(z)$

$$H_{\text{eff}} = E_0 + \Pi_- (V + VG_-(E_0)V + VG_-(E_0)VG_-(E_0)V + \dots) \Pi_- \quad (12)$$

where z is replaced with the unperturbed ground state energy E_0 . Thus we can only trust the result up to the first non-trivial order, the higher order terms may be less significant than the z -dependence of $\Sigma(z)$. Note that each single term in V creates two pairs of anti-aligned spins, and thus the perturbation series vanishes at odd orders. Let us consider the even orders:

1. $H_{\text{eff}}^{(2)} = \Pi_- VG_-(E_0)V\Pi_-$ gives a constant

$$H_{\text{eff}}^{(2)} = - \sum_{x\text{-link}} \frac{J_x^2}{4J_z} - \sum_{y\text{-link}} \frac{J_y^2}{4J_z} = -N \frac{J_x^2 + J_y^2}{4J_z}$$

Since it shifts the energy of the ground state manifold globally, we can proceed to the next order safely.

2. $H_{\text{eff}}^{(4)} = \Pi_- VG_-(E_0)VG_-(E_0)VG_-(E_0)V\Pi_-$ contains disconnected pieces that are twice of the second order terms. It also contains 24 connected pieces, each of which corresponds to flipping 4 spin pairs in a particular order. So we have

$$H_{\text{eff}}^{(4)} = \text{const} - \frac{J_x^2 J_y^2}{16J_z^3} \sum_p W_{p,\text{eff}}$$

where $W_{p,\text{eff}}$ is the effective representation of the integral of motion defined in (11).

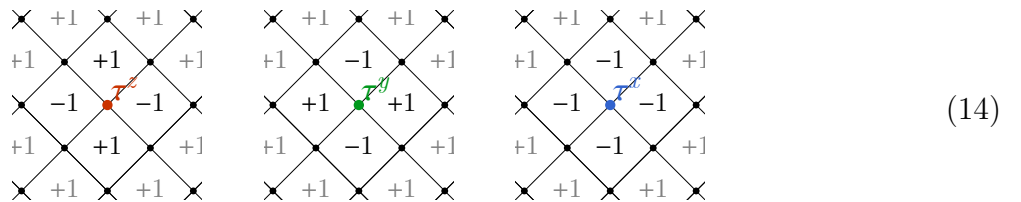
Thus, the effective Hamiltonian up to the constant term is

$$H_{\text{eff}} = - \frac{J_x^2 J_y^2}{16J_z^3} \sum_p W_{p,\text{eff}}, \quad W_{p,\text{eff}} = \tau_{\text{left}(p)}^y \tau_{\text{right}(p)}^y \tau_{\text{up}(p)}^z \tau_{\text{down}(p)}^z \quad (13)$$

3.2 Abelian anyons

In this subsection, we analyze the anyon content encoded in (13). The perturbed ground state is free of $\mathbb{Z}_2$ vortex, and each $\mathbb{Z}_2$ vortex has a constant energy cost. We will show that they form a set of anyons that is isomorphic to that of the toric code.

Let us first understand how these vortices can be created and moved by local operators. We can represent the model on a square lattice where τ spins are attached to the vertices. Let us examine how a single Pauli operator $\tau^{x,y,z}$ acts on the ground state, which is free of $\mathbb{Z}_2$ vortex. We can see that each Pauli operator has a different effect as drawn below: τ^z and τ^y creates a pair of $\mathbb{Z}_2$ vortices horizontally and vertically while τ^x creates four $\mathbb{Z}_2$ vortices simultaneously



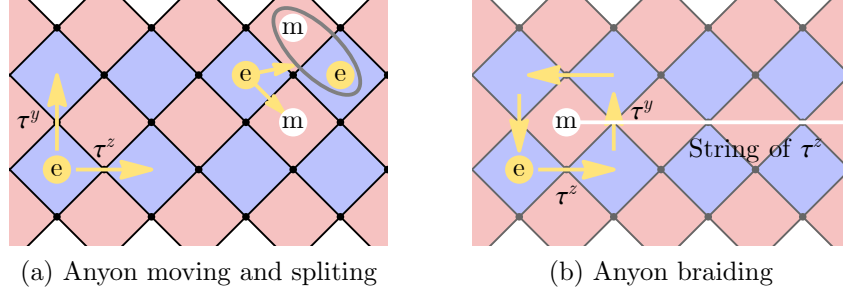


Figure 2: Abelian anyons in the gapped phase. There are e and m anyons living on the blue and red plaquette.

Consequently, given a single $\mathbb{Z}_2$ vortex, we move it horizontally by applying τ^z or vertically by applying τ^y . However, it cannot move along any diagonal direction (e.g. 45 degree) without creating other $\mathbb{Z}_2$ vortices.

Let us color the plaquettes in the odd and even rows by red and blue, respectively. Then the above analysis shows that $\mathbb{Z}_2$ vortices can only move between plaquettes with the same color. Therefore, it is meaningful to give them different labels. We call $\mathbb{Z}_2$ vortices on blue plaquettes the e -particles and on red plaquettes the m -particles. They are both boson and have the same energy, yet they cannot be transformed to each other without creating or annihilating more particles, as shown in Fig. 2 (a). The reason for the label e and m is that they are both boson and their mutual statistics has a minus sign

$$\begin{array}{c} \text{blue arrow} \\ \text{red arrow} \end{array} \text{ crossing } = - \begin{array}{c} \text{red arrow} \\ \text{blue arrow} \end{array} \quad (15)$$

One can check this by moving e around m on the lattice shown by Fig. 2 (b). Their bound state is a fermion f . We have the following fusion rules

$$e \times e = m \times m = f \times f = 1, \quad e \times m = f \quad (16)$$

Thus, the anyon content can be identified with the toric code model.

3.3 Weak symmetry breaking and gapless edge

However, this model is distinguished from the toric code by a strange feature: translation in the vertical direction by one lattice vector interchanges the e and m superselection sectors. This phenomenon is called the “weak symmetry breaking” in one-dimension by Kitaev.³ The above perturbative analysis applies to the regimes $J_x \gg J_{y,z}$ and $J_y \gg J_{x,z}$ as well and leads to the same anyon content. However, they differ by the weak symmetry breaking pattern, namely, e and m are interchanged by translations along different directions. This implies that the three perturbative regimes must be separated by transition even if we introduce new symmetry

³In the ordinary symmetry breaking, the symmetry transformation interchanges the different vacua. That is called the weak symmetry breaking in zero-dimension.

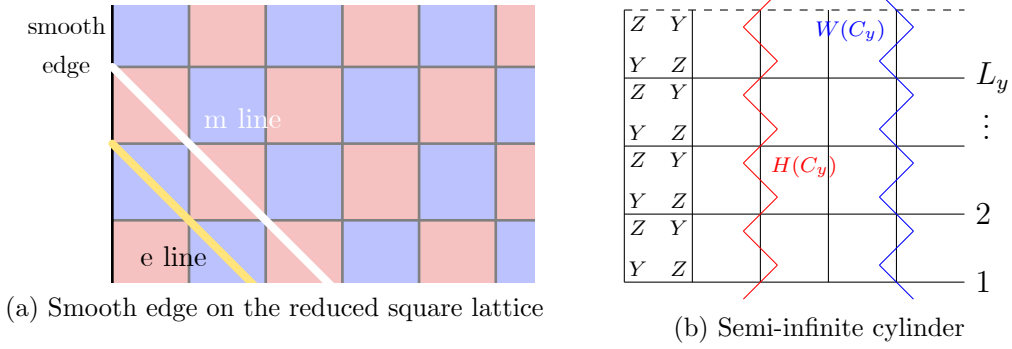


Figure 3: Smooth edge. The translation by one lattice vector along the edge relates e and m .

allowed terms in the Hamiltonian.⁴ In the exact calculation shown later, we will see that the three regimes are indeed separated by a gapless phase within the model.

Another implication of the weak symmetry breaking is on the edge physics. We first recall the edge physics of the toric code Hamiltonian. It has a *gapable edge* with either e or m condensed. In this case, the fact that translation can interchange e and m makes an important difference such that a certain type of edge is forced to be gapless.

There are two types of edges. Let us first consider the smooth edge shown in Fig.3(a). One can move the e and m particle to the edge without any energy cost, which implies that e and m are allowed to condense on the edge (which one actually condenses depends on the details of the perturbation). Let us assume that the translation in the vertical direction is not broken spontaneously. Since the translation interchanges e and m , they must condense or not *simultaneously*, which is different from the case of toric code. The non-trivial mutual statistics between them tells us that this is possible only when the edge is *gapless*.⁵ [read Levin's paper for better understanding]

Let us be more explicit on the above argument. We consider a semi-infinite cylinder with a smooth edge drawn in Fig.3(b) and the Hamiltonian takes the form (13). The dashed top line is identified with the bottom line. We call the horizontal (vertical) direction as x -(y -)direction, L_y denotes the length of the y -direction. The ground state is highly degenerate. First, note that the Wilson loop $W(C_y)$ and 't Hooft loop $H(C_y)$ along the y -direction commute with the Hamiltonian and thus gives rise to four superselection sectors.⁶ Second, note that there are operators on the edge that commute with the Hamiltonian and have non-trivial algebra

$$Z_{2\ell-1}Y_{2\ell}, \quad Z_{2\ell}Y_{2\ell+1}, \quad \ell = 1, 2, \dots, L_y \quad (17)$$

The operators in the two groups anti-commute with each other, and the product of the operators

⁴This is the analogue of the ordinary symmetry breaking. Phases with different symmetry breaking patterns are separated by either first order or continuous transition.

⁵We thank Ashvin Vishwanath for helpful discussion.

⁶A difference from the toric code is that the trajectories of e and m follow the zig-zag pattern shown in the figure. The color is to be consistent with the coloring in Fig.3

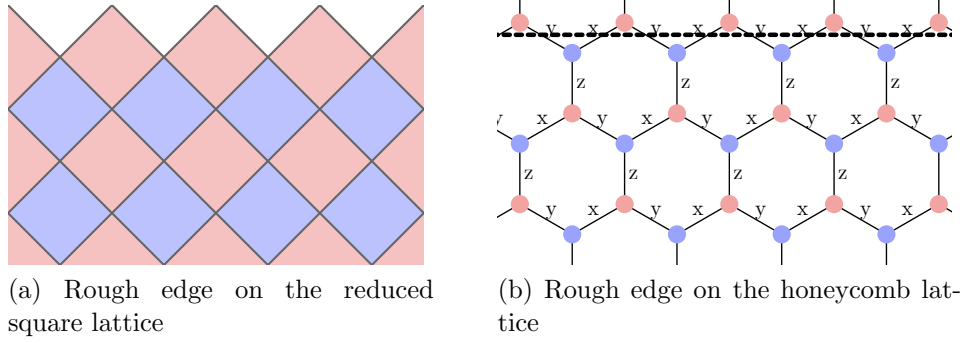


Figure 4: Rough edge. The translation by one lattice vector along the edge does not relate e and m in this case.

in two groups is exactly the Wilson loop and 't Hooft loop, i.e.

$$\prod_{\ell=1}^{L_y/2} Z_{2\ell-1} Y_{2\ell} = W(C_y), \quad \prod_{\ell=1}^{L_y/2} Z_{2\ell} Y_{2\ell+1} = H(C_y) \quad (18)$$

The ground state must have an extensive amount of degeneracy in order to support the algebra. Now, let us add *translationally invariant perturbation* to the Hamiltonian, which will generate an effective Hamiltonian H_{eff} for the ground state manifold and lift the degeneracy. To the leading order, H_{eff} only involves the above operators $Z_{\ell} Y_{\ell+1}$ in a translationally invariant fashion, i.e.

$$H_{\text{eff}} = -J \sum_{\ell=1}^{L_y/2} Z_{2\ell-1} Y_{2\ell} + Z_{2\ell} Y_{2\ell+1} + \dots \quad (19)$$

where $\dots$ represents the higher order terms. Note that the above Hamiltonian can be mapped to a critical transverse-field Ising model⁷

$$H_{\text{TFIM}} = -J \sum_{\ell=1}^{L_y/2} \sigma_{\ell}^x + \sigma_{\ell}^z \sigma_{\ell+1}^z + \dots \quad (20)$$

because the terms in the two Hamiltonian satisfy the same algebraic relation, which allows for a one-to-one correspondence between the spectrum. Under this mapping, the lattice translation of the two-dimensional system becomes the Kramer-Wannier duality of the effective description of the one-dimensional system. And requiring translation symmetry amounts requiring H_{TFIM} to be invariant under Kramer-Wannier duality, which implies that it must be gapless although we cannot determine the higher order terms. This is an explicit proof of the gaplessness of the smooth edge.

The other type is the rough edge shown in Fig.4. Notice that the translation along this edge maps anyons to the same type. Therefore, it is allowed to have either e or m condensed and becomes gapped.

⁷The Wilson loop and 't Hooft loop are mapped to the global $\mathbb{Z}_2$ charge and $\mathbb{Z}_2$ flux of the Ising model. This establishes a mapping between the superselection sector of the two-dimensional topological order and the charge-flux sector of the one-dimensional system.

4 Parton construction and the full phase diagram

We have to go beyond perturbation theory if we want to know the entire phase diagram of this model. Remarkably, this model can be solved exactly by using a fermionic representation. In this section, we will introduce this method and derive the full phase diagram.

4.1 Majorana fermion representation

We embed each spin into two fermionic modes, i.e., four Majorana operators, which are denoted by b^x, b^y, b^z, c . This embedding enlarges the dimension of the Hilbert space from two to four. We need to specify a convention of how the states and operators are embedded:

- The Hilbert space of the spin $\mathcal{M}$ is identified as a two-dimensional subspace of the four-dimensional fermion Fock space $\widetilde{\mathcal{M}}$ by this condition

$$|\xi\rangle \in \mathcal{M} \Leftrightarrow D|\xi\rangle = |\xi\rangle, \quad D = b^x b^y b^z c \quad (21)$$

We call $\mathcal{M}$ and $\widetilde{\mathcal{M}}$ the physical and extended Hilbert space. We can also interpret (21) as a gauge condition, namely, D is a gauge transformation and $\mathcal{M}$ is the corresponding gauge invariant subspace. The reason for this terminology will be clear later.

- The Pauli operators $\sigma^{x,y,z}$ can be represented by operators $\widetilde{\sigma}^{x,y,z}$ acting on $\widetilde{\mathcal{M}}$. Such a representation must satisfy two conditions: (1) $\widetilde{\sigma}^{x,y,z}$ preserves $\mathcal{M}$, i.e. $[\widetilde{\sigma}^\alpha, D] = 0$; (2) they obey the same algebra as $\sigma^{x,y,z}$ when restricted to $\mathcal{M}$. The representation is not unique and we pick the following choice

$$\begin{array}{c} b_z \\ \bullet \\ c \\ \bullet \\ b_x \quad \bullet \quad b_y \end{array} \quad \widetilde{\sigma}^x = ib^x c, \quad \widetilde{\sigma}^y = ib^y c, \quad \widetilde{\sigma}^z = ib^z c \quad (22)$$

Notice that $\widetilde{\sigma}^x \widetilde{\sigma}^y \widetilde{\sigma}^z = iD$, which implies that they satisfies the Pauli algebra in the subspace $\mathcal{M}$ ($D = 1$).

Now, let us apply the above general embedding procedure to the Hamiltonian (6). Each term becomes $\sigma_j^\alpha \sigma_k^\alpha \rightarrow i(-ib_j^\alpha b_k^\alpha) c_j c_k$. We denote the operator in parentheses by u_{jk} , which is Hermitian and $\mathbb{Z}_2$ -valued $u_{jk} = \pm 1$. Thus, we have

$$H \rightarrow \widetilde{H} = \frac{i}{4} \sum_{\langle jk \rangle} \widehat{A}_{jk} c_j c_k, \quad \widehat{A}_{jk} = 2J_{\alpha_{jk}} \widehat{u}_{jk}, \quad \widehat{u}_{jk} = ib_j^{\alpha_{jk}} b_k^{\alpha_{jk}} \quad (23)$$

We use the notation $\widetilde{H}$ to emphasize that it acts on the enlarged many-body Hilbert space $\widetilde{\mathcal{H}}$. It respect the gauge condition by construction $[\widetilde{H}, D] = 0$. In the following, we first analyze $\widetilde{H}$ in the extended Hilbert space $\widetilde{\mathcal{H}}$ and then consider the restriction to $\mathcal{H}$.

The extended Hamiltonian (23) also has a set of symmetries. It has translation and time-reversal symmetries that are inherited from H , where the time reversal can be defined as⁸

$$T b^\alpha T^{-1} = b^\alpha, \quad T c T^{-1} = c \quad (24)$$

⁸One can use a different definition long as it can satisfy the physical requirement $T \sigma^\alpha T^{-1} = -\sigma^\alpha$.

Remarkable, (23) also has an extensive number of *local* symmetries. Notice that $\hat{u}_{jk}$ commute with the Hamiltonian and with each other. Therefore, we can divide the extended Hilbert space $\mathcal{H}$ into common eigenspaces of $\hat{u}_{jk}$

$$\tilde{\mathcal{H}} = \bigoplus_{\{u\}} \tilde{\mathcal{H}}_{\{u\}} \quad (25)$$

In each subspace $\tilde{\mathcal{H}}_{\{u\}}$, $\hat{u}_{jk}$ becomes numbers, which results in a free fermion Hamiltonian $\tilde{H}_u = \frac{i}{4} \sum_{\langle jk \rangle} A_{jk} c_j c_k$. The ground state can be solve exactly, which we denote by $|\tilde{\Psi}_u\rangle$.

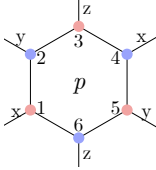
Then we impose the gauge condition (21). However, the subspace $\tilde{\mathcal{H}}_{\{u\}}$ is not gauge invariant. Applying D_j changes the value of u_{jk} on the three links connecting the vertex j . Therefore, we can identify $\tilde{\mathcal{H}}_{\{u\}}$ related by gauge transformation to get the physical Hilbert space and a corresponding decomposition. More explicitly, let w denote the gauge equivalent class of u , then we have

$$\tilde{\mathcal{H}} = \bigoplus_{\{u\}} \tilde{\mathcal{H}}_{\{u\}} \rightarrow \mathcal{H} = \bigoplus_w \mathcal{H}_w, \quad |\Psi_w\rangle = \prod_j \frac{1 + D_j}{2} |\tilde{\Psi}_u\rangle \in \mathcal{H}_w \quad (26)$$

It follows from $[D, \tilde{H}] = 0$ that $|\Psi_w\rangle$ is a gauge invariant ground state in the gauge equivalent class w . The gauge equivalent class w is nothing but the product of u_{jk} around a plaquette

$$w_p = - \prod_{\langle jk \rangle} u_{jk} \quad (jk) \text{ arranged in the counterclockwise order} \quad (27)$$

One can check that it is gauge invariant. Actually, this decomposition of $\mathcal{H}$ by gauge equivalent classes is exactly the same as the decomposition by the local integrals of motion W_p that we obtained before (see (9)). This can be verified by comparing the operator W_p for the local integral of motion and the operator $\tilde{W}_p$ that corresponds to the gauge equivalent classes

$$\begin{aligned} \tilde{W}_p &= - \prod_{\langle jk \rangle} \hat{u}_{jk} \\ &= - (ib_1^y b_6^y) (ib_6^x b_5^x) (ib_5^z b_4^z) (ib_4^y b_3^y) (ib_3^x b_2^x) (ib_2^z b_1^z) \\ &= (ib_1^y b_1^z) (ib_2^z b_2^x) (ib_3^x b_3^y) (ib_4^y b_4^z) (ib_5^z b_5^x) (ib_6^x b_6^y) \\ &= (ib_1^x c_1) (ib_2^y c_2) (ib_3^z c_3) (ib_4^x c_4) (ib_5^y c_5) (ib_6^z c_6) \\ &= \tilde{\sigma}_1^x \tilde{\sigma}_2^y \tilde{\sigma}_3^z \tilde{\sigma}_4^x \tilde{\sigma}_5^y \tilde{\sigma}_6^z \\ &= W_p \end{aligned}$$


In the first step, we use the definition of $\hat{u}_{jk}$. In the second step, we rearrange the operators. In the third step, we use the condition $D = 1$ in the physical subspace. In the rest steps, we use the definition of $\tilde{\sigma}^\alpha$ and that we are in the physical subspace.

We have kept using the term “gauge”. Let us make the connection to gauge theory more clearly. The model in the fermion representation can be regarded as a $\mathbb{Z}_2$ gauge theory:

- The link variable $\hat{u}_{jk}$ is the lattice gauge field that is coupled to Majorana fermions on the vertices. And D_j is the local gauge transformation. The physical Hilbert space $\mathcal{H}$ corresponds to the gauge invariant subspace and we can decompose the Hilbert space $\mathcal{H}$ into sectors $\mathcal{H}_w$ labeled by the gauge flux w .

- The solvability of this model comes from the fact that the gauge field does not have dynamics so that we can choose a sector $\mathcal{H}_w$ and solve a free-fermion Hamiltonian after fixing a gauge.

In this gauge theory language, let us give an incomplete dictionary between the physical operators of spin and gauge invariant operators of the gauge theory:

- $\mathbb{Z}_2$ vortex corresponds to the Wilson loop operator

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z \leftrightarrow \widetilde{W}_p = - \prod_{\langle jk \rangle \text{ c.c.w.}} \widehat{u}_{jk} \quad (28)$$

- The product of the two-spin couplings K_{jk} that appears in the Hamiltonian (6) corresponds to fermionic path operator

$$K_{j_n j_{n-1}} \cdots K_{j_1 j_0} \leftrightarrow W(j_0, \cdots, j_n) = \left(\prod_{s=1}^n -i \widehat{u}_{j_s, j_{s-1}} \right) c_n c_0 \quad (29)$$

When the path is closed $j_n = j_0$, this correspondence reduces to (28).

- A single Pauli operator corresponds to the monopole creating/annihilation operator

$$\sigma_j^\alpha \rightarrow M_j^\alpha = i b_j^\alpha c_j \quad (30)$$

Note that the right hand side cannot be written in terms of c and u_{jk} . This is a generic feature of monopole operators. Applying the monopole operators changes the vortex configuration and maps states between different sectors $\mathcal{H}_w$.

4.2 The spectrum and the phase diagram

Now, we study the spectrum of the model using the Majorana fermion representation. The strategy is to choose a subspace $\mathcal{H}_w$ that has a definite gauge flux configuration w and fix a gauge to diagonalize a free-fermion Hamiltonian $H(A)$.

First, we notice that the spectrum is independent of the sign of $J_{x,y,z}$ because they can always be compensated by gauge transformation on u_{jk} . Therefore, without loss of generality, we assume $J_{x,y,z} \geq 0$.

Then, we determine the phase diagram by calculating the low-energy spectrum. We need to find a vortex configuration that minimizes the ground state energy. It turns out that the energy minimum is achieved by the vortex-free configuration $w_p = 1$.⁹ A single vortex or even a pair of nearest-neighbor vortices has a finite energy cost. This statement follows a theorem proved by Lieb. In the following, we choose the gauge $u_{jk} = 1$ to have a translationally invariant Hamiltonian and the diagonalization becomes simple.

⁹However, the first excited state does not have to belong to the vortex-free sector. We will see that the first excited state in the gapped phase corresponds to creating vortex.

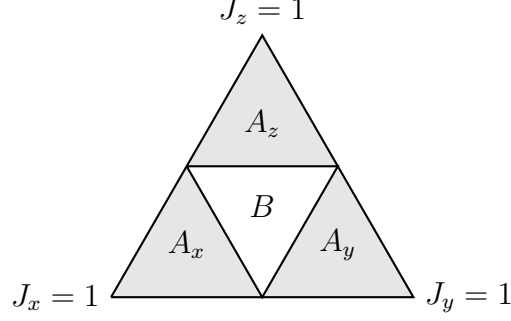
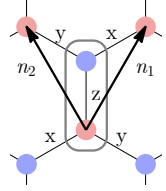


Figure 5: Phase diagram of the model. The triangle is the section of the positive octant $J_{x,y,z} \geq 1$ by the plane $J_x + J_y + J_z = 1$. The three A phases are gapped and support abelian anyons that are isomorphic to that of the toric code. The B phase is gapless.

The honeycomb lattice has two sublattice. We choose the following unit cell and basis of the translation group. The Hamiltonian in the momentum space is



$$H_{\text{vortex-free}} = \frac{1}{2} \sum_q \begin{pmatrix} a_{-q,r} & a_{-q,b} \end{pmatrix} i\tilde{A}(q) \begin{pmatrix} a_{q,r} \\ a_{q,b} \end{pmatrix} \quad (31)$$

$$i\tilde{A}(q) = \begin{pmatrix} 0 & if(q) \\ -if(q)^* & 0 \end{pmatrix}, \quad f(q) = 2(J_x e^{iq \cdot n_1} + J_y e^{iq \cdot n_2} + J_z)$$

Here, the subscripts r, b label the red and blue sublattice. We choose the convention that $u_{jk} = 1$ for $j \in \text{blue}, k \in \text{red}$. And a_q is the momentum representation of the Majorana fermion operator

$$a_{q,r/b} = \frac{1}{\sqrt{2N}} \sum_s e^{-iq \cdot r_s} c_{s,r/b} \quad (32)$$

that satisfies the canonical anti-commutation relation $\{a_{p,\alpha}, a_{q,\beta}^\dagger\} = \delta_{\alpha\beta} \delta_{p,q}$. The single particle spectrum is thus given by

$$\varepsilon(q) = \pm |f(q)| \quad (33)$$

An important property of this single-particle spectrum is whether it is gapless. This is determined by whether the following equation has solution for some q

$$J_x e^{iq \cdot n_1} + J_y e^{iq \cdot n_2} + J_z = 0 \quad (34)$$

This is possible only if $J_{x,y,z}$ satisfies

$$J_x \leq J_y + J_z, \quad J_y \leq J_z + J_x, \quad J_z \leq J_x + J_y \quad (35)$$

Thus, we can obtain the full phase diagram, which is shown in Fig. 5.

Gapped phases There are three gapped phases A_x , A_y and A_z in the three regions $J_x \geq J_y + J_z$, $J_y \geq J_z + J_x$ and $J_z \geq J_x + J_y$, respectively, which are what we have analyzed in Sec. 3. The exact calculation of the spectrum confirms our perturbative analysis. Each of the

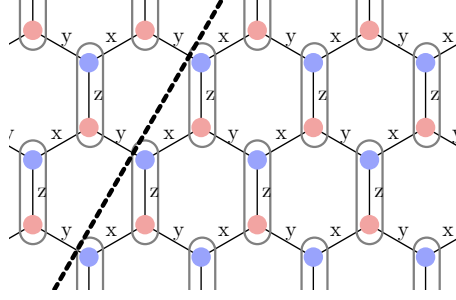


Figure 6: Smooth edge on the honeycomb lattice. We consider the limit $J_z \gg 1$ and the gray circles denote the pairing between fermions to the leading order in J_z .

three phases supports Abelian anyons $\{1, e, m, f = em\}$ which are isomorphic to the anyon content of toric code. The three phases are further distinguished from each other by the weak symmetry breaking pattern, i.e., how the translation acts on the superselection sectors.

For the gapped phase, one may wonder the relation between the fermion f and the Majorana fermion c . They are in the same superselection sector because they are both fermions and have the same braiding rules with the $\mathbb{Z}_2$ vortices e and m .¹⁰ However, they corresponds to completely different states with different energies. For simplicity, let us consider the limit $J_z \gg J_{x,y}$:

- The state that contains (a pair of) Majorana fermion c is in the vortex-free sector, which has an energy of order J_z .
- On the other hand, the state that contains (a pair of) fermion f is in sectors that have $\mathbb{Z}_2$ vortex, which has an energy of order $\frac{J_x^2 J_y^2}{J_z^3} \ll J_z$.

Due to the conservation of W_p , the two types of fermions cannot decay into each other.

It is also interesting to consider the smooth edge in the fermion picture, which is shown in Fig. 6. To the leading order, pairs of fermions across the z -links are paired up. The y -links and x -links are perturbations. In the whole process, the system in a given sector $\tilde{\mathcal{H}}_{\{u\}}$ remains gapped. However, we also prove that the edge is gapless in Sec. 3.3. How to reconcile this contradiction? The reason is that the gapless excitations on the edge have nothing to do with the Majorana fermions c . They are associated to e, m and f , which are created by (monopole) operators that act across different flux sectors.

Gapless phase When $J_{x,y,z}$ are at the intermediate parameter regime, the system enters a gapless B phase and cannot have anyons on its own. A natural question is what perturbation can open a gap and also a window for anyons. The easiest way is to add diagonal terms to the matrix $\tilde{A}(q)$. Let us show that this is forbidden by the time-reversal symmetry.

The time reversal T defined in (24) changes the sign of $\hat{u}_{jk} = ib_j^{\alpha_{jk}} b_k^{\alpha_{jk}}$ for all links (jk) . The magnetic flux, which involves an even number of $\hat{u}_{jk}$, does not change. And we can compensate it by a gauge transformation, e.g., a gauge transformation on all the blue/red sites in Fig. 1.¹¹

¹⁰For the Majorana fermion c , the braiding rule with e and m is the AB phase.

¹¹Therefore, this is a feature of bipartite lattice. A triangular lattice will be quite different.

Thus, we have a physically equivalent time-reversal operator T' defined as

$$T' \hat{u}_{jk} T'^{-1} = \hat{u}_{jk}, \quad T' c_j T'^{-1} = \begin{cases} c_j & \text{red sites} \\ -c_j & \text{blue sites} \end{cases} \quad (36)$$

Consequently, a T' -invariant perturbation to the fermion Hamiltonian $\tilde{H}$ (in a fixed gauge) cannot contain any term $ic_j c_k$ where j, k belong to the same sublattice. The perturbed matrix $\tilde{A}(q)$ still has zero diagonal elements and has gapless points.

5 Magnetic field and non-Abelian anyons

In order to make the gapless B phase more interesting for our purpose here (to find anyons), we have to break the time reversal symmetry. We will now show that the following perturbation

$$V = - \sum_j (h_x \sigma_j^x + h_y \sigma_j^y + h_z \sigma_j^z) \quad (37)$$

does open a gap and further leads to non-abelian anyons. Let us summarize the results below before diving into the details:

- The bulk c Majorana fermion has a spectral Chern number $\nu = \pm 1$, which implies that the π -flux carries unpaired Majorana modes. This is similar to the $p + ip$ superconductor (reviewed in Appendix C). The only difference is that the $\mathbb{Z}_2$ gauge field is dynamical and the π -flux is a part of the spectrum.
- The bulk has three superselection sectors $1, \sigma, \varepsilon$ corresponding to the vacuum, $\mathbb{Z}_2$ -flux and a fermion. Their fusion rules are

$$\varepsilon \times \varepsilon = 1, \quad \varepsilon \times \sigma = \sigma, \quad \sigma \times \sigma = 1 + \varepsilon \quad (38)$$

The fusion rule of σ contains multiple channels, which is the defining feature of non-abelian anyons. The physical meaning is the following. Before the fusion, there are two practically indistinguishable states with two σ -particles, i.e., no local measurement can tell them apart. The fusion can be regarded as a measurement that tells them apart.¹²

- The edge has a single chiral Majorana fermion, which is also and must be gapless. What is the difference compared with the gapless edge in the A phase? For brevity, denote them by A and B edges. The A edge is non-chiral and corresponds to the entire Ising CFT. The B edge is chiral and has a chiral central charge $c_- = 1/2$, which leads to a physical difference: the B edge has a quantized thermal Hall conductance:

$$\lim_{T \rightarrow 0} \frac{\kappa_{xy}}{T} = \frac{\pi}{6} c_- \quad (39)$$

Analogous to that quantum Hall effect is associated to the chiral anomaly of the edge theory (the contact term in the jj OPE), this thermal Hall effect is also associated to an anomaly but a different one, the perturbative gravitational anomaly (the contact term in the TT OPE).

¹²Another method is to introduce a third particle and let it braid with the two σ -particles.

5.1 Effective Hamiltonian and the spectral Chern number

In this subsection, we argue the form of the effective Hamiltonian and show that it has a non-zero spectral Chern number. We will also explain why it implies that a π -flux carries an unpaired Majorana modes. For simplicity, we assume $J_x = J_y = J_z = J$.

Let us apply the Green's function formalism to determine the effective Hamiltonian for the vortex-free sector. Note that each term in V are the monopole operator in the gauge theory. The first order term $H_{\text{eff}}^{(1)} \sim \Pi$ vanishes. Although the second order term $H_{\text{eff}}^{(2)}$ does not vanish, it preserves the time-reversal symmetry. And we must consider the third order term

$$H_{\text{eff}}^{(3)} = \Pi_- V G_-(z) V G_-(z) V \Pi_-$$

In principle, we have to choose appropriate z to include the second order correction and calculate G_- to obtain an exact answer for $H_{\text{eff}}^{(3)}$. However, it is rather difficult and also unnecessary. It suffices to using a qualitative argument to derive the possible form of $H_{\text{eff}}^{(3)}$ and showing that it can open a gap.

Let us assume that all intermediate states involve in the calculation have the same energy $\Delta E \sim J$ above the ground state. Then the effective Hamiltonian becomes

$$H_{\text{eff}}^{(3)} \sim -\frac{h_x h_y h_z}{J^2} \sum_{j,k,l} \sigma_j^x \sigma_k^y \sigma_l^z \quad (40)$$

where j, k, l are (1) either three adjacent sites (2) or the three blue(red) sites connecting to the same red(blue) sites. Note that j, k, l have to be arranged appropriately so that the product commutes with all W_p 's, which provides strong constraints on the allowed terms. The first case corresponds to a next-nearest neighbor hopping term for the Majorana fermion, which contributes to the diagonal term of $\tilde{A}(q)$. The second case corresponds to a four-fermion interaction and thus does not directly influence the spectrum. Thus, we arrived at the following effective Hamiltonian

$$i\tilde{A}(q) = \begin{pmatrix} \Delta(q) & if(q) \\ -if(q)^* & -\Delta(q) \end{pmatrix}, \quad \begin{aligned} f(q) &= 2J(e^{iq \cdot n_1} + e^{iq \cdot n_2} + 1) \\ \Delta(q) &= 4\kappa(\sin q \cdot n_1 + \sin q \cdot n_2 + \sin q \cdot (n_2 - n_1)) \end{aligned} \quad (41)$$

The exact form of $f(q)$ and $\Delta(q)$ is not essential. What is more important is that, by the time-reversal symmetry, $f(q)$ has two zeros at q^* and $-q^*$, whereas $\Delta(q)$, which breaks the time-reversal and thus is an odd function of q , satisfies $\Delta(q^*) = -\Delta(-q^*) \neq 0$. It not only opens a gap but also implies a non-zero spectral Chern number.

5.2 Edge mode, thermal transport and unpaired Majorana fermion

The non-zero spectral Chern number in the bulk implies a chiral Majorana fermion on the edge. One simple derivation is to consider the Jackiw-Rebbi setup. The fermion spectrum in the vortex-free sector contains two massive Dirac fermion at $\pm q^*$. Let us focus on one of them, say q^* , and write the corresponding low-energy Hamiltonian as

$$i\tilde{A}(q) \approx \begin{pmatrix} \Delta_0 & iv_F(q_x + iq_y) \\ -iv_F(q_x - iq_y) & -\Delta_0 \end{pmatrix}$$

where $q_{x,y}$ denotes the momentum deviation from q^* and we have redefined the x, y directions such that the off-diagonal elements take simple forms, $v_F \sim J$ has the meaning of velocity. Then we gradually tune Δ_0 in space so that it remains constant in the y -direction and interpolates between $-\Delta_0$ and Δ_0 in the x -direction as follows

$$\Delta(x) = \begin{cases} -\Delta_0 & x \rightarrow -\infty \\ \Delta_0 & x \rightarrow +\infty \end{cases}, \quad \Delta_0 > 0$$

Namely, we create an interface at $x = 0$ which separates the original system at $x > 0$ and another system with a zero Chern number at $x < 0$. The Hamiltonian in the real-space can be written as

$$iA = \begin{pmatrix} \Delta(x) & iv_F(-i\partial_x + iq_y) \\ -iv_F(-i\partial_x + iq_y) & -\Delta(x) \end{pmatrix} \quad (42)$$

which yields the following solution with energy $E_{q_y} = v_F q_y$

$$e^{iq_y y} \exp\left(\frac{1}{v_F} \int_0^x \Delta(x') dx'\right) \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (43)$$

It describes a chiral Majorana fermion which is localized near $x = 0$ and move along the y -direction. The corresponding low-energy Hamiltonian is

$$H = \frac{i}{4} \int dy \eta(y) (-iv_F \partial_y) \eta(y) \quad (44)$$

The chiral Majorana fermion does not carry charges and cannot have any effect on electric responses. However, it carries energy and can contribute to thermal responses, the thermal Hall effect. Namely, the edge prepared at a finite temperature T can carry nonzero energy current

$$I_E = \frac{\pi}{12} c_- T^2 \quad (45)$$

It is proportional to T^2 by dimensional analysis, the prefactor $\pi/12$ is a convention to make a connection conformal field theory. The non-trivial part is c_- , which is quantized in this case.¹³

The existence of the chiral Majorana edge mode also provides a way to understand the unpaired Majorana zero mode attached to the vortex. Let us consider a disk geometry where a single chiral Majorana mode moves along the edge. In the vortex free sector, the edge mode has an *anti-periodic* boundary condition.¹⁴ If we insert a single vortex in the bulk, it has a branch cut that terminates on the edge. Accordingly, the boundary condition on the edge must be changed to the periodic one. This brings a Majorana zero mode to the boundary. The full system has to have a well-defined Hilbert space, which implies that there is also an unpaired Majorana mode at the vortex.

¹³Strictly speaking, any two-dimensional topological order is trivial at finite temperature. Therefore, this formula should be understood as a result of the linear response.

¹⁴One way of proving this statement is to notice that the local frame for the edge mode rotates and gives a Berry phase. Here, we give a different argument. The total Hilbert space dimension must be an integer. However, the edge with a periodic boundary condition carries a zero mode, which must be compensated by another one in the bulk. This is impossible if the bulk preserves translational and rotational symmetry. Therefore, the edge has to have the anti-periodic boundary condition.

5.3 Fusion and braiding

A Perturbation in the Green's function formalism

In this section, we will introduce perturbation theory in the Green's function formalism. Our discussion follows [3]. Since this is a useful technique, we will try to keep the presentation as general as possible. We will also compare it with the Schrieffer–Wolff transformation.

Let us introduce some notation. The full Hamiltonian is $\tilde{H} = H + V$. We refer to H as the unperturbed Hamiltonian, which has a ground state manifold $\mathcal{L}_-$ near zero energy, whereas the excited state manifold $\mathcal{L}_+$ is separated by a gap Δ in the spectrum. Namely, the Hilbert space has a natural decomposition with respect to the spectrum

$$\mathcal{H} = \mathcal{L}_- \oplus \mathcal{L}_+$$

We refer to V as the perturbation, which satisfies $\Delta \gg \|V\|$.¹⁵ Therefore, we expect the gap to maintain in the spectrum of $\tilde{H}$ and we can do a similar decomposition

$$\mathcal{H} = \tilde{\mathcal{L}}_- \oplus \tilde{\mathcal{L}}_+$$

The goal of perturbation is to approximate the low-energy spectrum, i.e., eigenvalues of $\tilde{H}|_{\tilde{\mathcal{L}}_-}$. We will mark everything related to the full Hamiltonian with a tilde.

Let us introduce the Green's function (resolvents) of $\tilde{H}$, which is a meromorphic operator-valued function of the complex variable z with poles corresponding to the eigenvalues of $\tilde{H}$

$$\tilde{G}(z) = \frac{1}{z - \tilde{H}} = \sum_i \frac{1}{z - \tilde{\lambda}_i} |\tilde{\psi}_i\rangle \langle \tilde{\psi}_i| \quad (46)$$

Similarly, we define the Green's function of H as $G(z) = (z - H)^{-1}$. The usefulness comes from the fact that poles can be preserved under appropriate projection.

Let us define Π_- and Π_+ as the projector onto the ground state manifold $\mathcal{L}_-$ and excited states $\mathcal{L}_+$ of H , respectively. Any operator X acting on $\mathcal{H}$ can be written in the following block form

$$X = \begin{pmatrix} X_{++} & X_{+-} \\ X_{-+} & X_{--} \end{pmatrix}, \quad X_{ab} = \Pi_a X \Pi_b$$

Note that H and $G(z)$ are diagonal in this decomposition, and we write them as

$$H = \begin{pmatrix} H_+ & 0 \\ 0 & H_- \end{pmatrix}, \quad G(z) = \begin{pmatrix} G_+(z) & 0 \\ 0 & G_-(z) \end{pmatrix}$$

The perturbation V and full Hamiltonian have off-diagonal elements. What we are most interested is $\tilde{G}_{--}(z)$, the projection of $\tilde{G}(z)$ onto the ground state manifold $\mathcal{L}_-$. Physically, we expect that $\tilde{\mathcal{L}}_-$ and $\mathcal{L}_-$ have large overlap with each other so that all the eigenvalues of $\tilde{H}|_{\mathcal{L}_-}$

¹⁵Notice that any many-body system cannot satisfy this condition because the gap Δ is constant while $\|V\|$ is extensive in the system size. Fortunately, when *locality* plays an important role, the formula that we are going to derive still holds. For local many-body systems, the more appropriate condition is $\Delta \gg \|V_x\|$, where $V_x \subset V$ is the local term supported around position x .

should appear as poles of $\tilde{G}_{--}(z)$. Namely, the projection does not lose any information of the low-energy spectrum. This is indeed true and summarized by the following lemma. The proof can be found in [3].¹⁶

Lemma A.1. *Let $\tilde{\lambda}$ denote an eigenvalue of $\tilde{H}|_{\tilde{\mathcal{E}}_-}$ with multiplicity $m \geq 1$. Then around $\tilde{\lambda}$, $G_{--}(z)$ is of the form $(z - \tilde{\lambda})^{-1}A + \mathcal{O}(1)$, where A is a rank m operator.*

Therefore, it suffices to study the poles of $\tilde{G}_{--}(z)$. Let us introduce the *self-energy* operator¹⁷

$$G_{--}(z) = \frac{1}{z - \Sigma_-(z)} \quad \text{or} \quad \Sigma_-(z) = z - G_{--}^{-1}(z) \quad (47)$$

which translates the poles of $G_{--}(z)$ into eigenvalues of $\Sigma_-(z)$. The convenience of using Σ_- partially comes from that it has a natural series expansion. Let us express $\tilde{G}$ in terms of G

$$\tilde{G} = (G^{-1} - V)^{-1} = \begin{pmatrix} G_+^{-1} - V_{++} & -V_{+-} \\ -V_{-+} & G_-^{-1} - V_{--} \end{pmatrix}^{-1}$$

Using the block matrix identity, we can find that

$$\tilde{G}_{--}^{-1} = G_-^{-1} - V_{--} - V_{-+}(G_+^{-1} - V_{++})^{-1}V_{+-}$$

Finally, we obtain the following series expansion

$$\Sigma_-(z) = V_{--} + V_{-+}G_+V_{+-} + V_{-+}G_+V_{++}G_+V_{+-} + \dots \quad (48)$$

Notice that the convergence of this series expansion relies on the condition $\|V\|/\Delta \ll 1$.¹⁸

From the definition (47), we can see that $\Sigma_-(z)$ plays the role of a Hamiltonian for the projected Green's function $\tilde{G}_{--}$. However, $\Sigma_-(z)$ has a z -dependence and is not a fixed Hamiltonian. More explicitly, an actual eigenvalue λ in the spectrum is related to the eigenvalue of Σ_- by

$$\Sigma_-(\lambda)|\psi\rangle = \lambda|\psi\rangle$$

Nonetheless, we can choose an effective Hamiltonian H_{eff} that approximates $\Sigma_-(z)$ in the range we are interested in. Ref. [3] proves that $\|H_{\text{eff}} - \Sigma_-(z)\| < \varepsilon$ for all z in a certain low-energy range guarantees an ε -close approximation to the actual eigenvalues of $\tilde{H}$ in that range.¹⁹ One common choice of H_{eff} is to use the expansion (48) and replace z by an appropriate constant (usually the unperturbed ground state energy).

¹⁶Strictly speaking, the proof assumes quantum mechanical systems (i.e. $\Delta \gg \|V\|$). Since the conclusion makes sense physically, we believe that generalization to many-body systems is possible.

¹⁷Comparing with the more commonly used self-energy, this definition includes a H_- term for notational convenience.

¹⁸For a generic many-body system of size N , we expect the k -th term in the expansion to be of order $\mathcal{O}(N^k)$ and the series expansion is always divergent if $N \gg 1$. However, when we consider local systems, locality can provide strong constraints. It turns out that all terms in the expansion is of order $\mathcal{O}(N)$, and the series is convergent if $\|V_x\|/\Delta \ll 1$.

¹⁹That proves relies on $\Delta > \|V\|/2$ and does not work directly in many-body systems. But we expect that a generalization is possible.

Let us recall the Schrieffer–Wolff transformation. The effective Hamiltonian is defined straightforwardly as

$$H_{\text{eff}} = \Pi_- U (H + V) U^\dagger \Pi_- \quad (49)$$

where U is a unitary transformation from $\tilde{\mathcal{L}}_-$ to $\mathcal{L}_-$. The existence of U is guaranteed by the condition $\|V\| \ll \Delta$ and generalization of this condition to many-body systems is also known. The construction of U and H_{eff} can be done systematically.

In the Green’s function approach, the effective Hamiltonian is defined as an approximation to the self-energy operator. However, there is no systematic way of finding the effective Hamiltonian and controlling the error. This is the main disadvantage of this approach compared with the Schrieffer–Wolff transformation. If the interesting physics appears at the first non-vanishing order, this Green’s function approach is more efficient.

B Another fermion parton construction

Parton construction is a systematic method of constructing more interesting many-body ground states and the corresponding low-energy theories, that are beyond the scope of ordinary symmetry-breaking states. It was developed around 1980s and has been widely applied to the study of spin liquid, quantum Hall physics, etc. The Majorana fermion representation introduced in Sec. 4.1 is one particular parton construction, which not only greatly simplifies the problem but also leads to interesting topologically ordered states. In this section, we will introduce a different parton construction, *the slave fermion*. We will see that parton construction generally cannot be solved exactly. It can be regarded as a novel mean-field method that can hint the *possible phases*. However, the calculation is usually poorly under control. The discussion follows Michael Levin’s UQM lecture [4].

B.1 Slave fermion: generality

B.2 Examples

C Review of $p + ip$ superconductor

We consider (2+1)-dimensional spinless fermions moving and assume that the system is already in a superconducting phase so that we can use the following BCS Hamiltonian,

$$H = \int d^2x \, c^\dagger(x) \left(-\frac{\partial^2}{2m} - \mu \right) c(x) + \sum_{\alpha=x,y} \int d^2x \, (\Delta_\alpha c^\dagger(x) \partial_\alpha c^\dagger(x) + h.c.), \quad (50)$$

where we only consider the p-wave pairing component. Only the relative phase between Δ_x and Δ_y is physically meaningful. We can also write down a lattice version for the Hamiltonian above, e.g., fermion hopping on a square lattice,

$$H = -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + h.c.) - \sum_i (\mu - 4t) c_i^\dagger c_i + \sum_i \sum_{\alpha=x,y} (\Delta_\alpha c_i^\dagger c_{i+\alpha}^\dagger + h.c.), \quad (51)$$

where $\langle ij \rangle$ denotes nearest-neighbor sites.

The ground state is determined by self-consistently solving the gap equations and number equation. Here we only quote the result: on a plane and at low enough temperature, the ground state is in a chiral phase, i.e. $\Delta_x = i\Delta_y \neq 0$ or $\Delta_x = -i\Delta_y \neq 0$, which are called $p + ip$ and $p - ip$ respectively. The two phases are connected by time reversal transformation thus have similar physical properties. From now on, we focus only on the $p + ip$ phase. We will primarily use the lattice Hamiltonian to avoid the UV divergence and go to the continuum by taking $k \rightarrow 0$ whenever we want. Thus, we consider the following Hamiltonian,

$$\begin{aligned}
H_{p+ip} = & -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + h.c.) - \sum_i (\mu - 4t) c_i^\dagger c_i \\
& + \sum_i (\Delta c_i^\dagger c_{i+x}^\dagger + h.c.) + (-i\Delta c_i^\dagger c_{i+y}^\dagger + h.c.).
\end{aligned} \tag{52}$$

In addition to translation symmetry, this Hamiltonian has another three symmetries: fermion parity $\mathbb{Z}_2^F$, time-reversal $T : c_i \rightarrow c_i$ and particle-hole. Note that the Hamiltonian is actually covariant under time-reversal symmetry $TH_{p+ip}[\Delta]T^{-1} = H_{p-ip}[\Delta^*]$.

References

- [1] A. Kitaev, *Anyons in an exactly solved model and beyond*, *Annals Phys.* **321** (2006) 2 [[cond-mat/0506438](#)].
- [2] A. Kitaev and C. Laumann, *Topological phases and quantum computation*, *arXiv e-prints* (2009) arXiv:0904.2771 [[0904.2771](#)].
- [3] J. Kempe, A. Kitaev and O. Regev, *The Complexity of the Local Hamiltonian Problem*, *arXiv e-prints* (2004) quant [[quant-ph/0406180](#)].
- [4] L. Michael, *Parton, lectures at ultra-quantum matter summer school*, .